



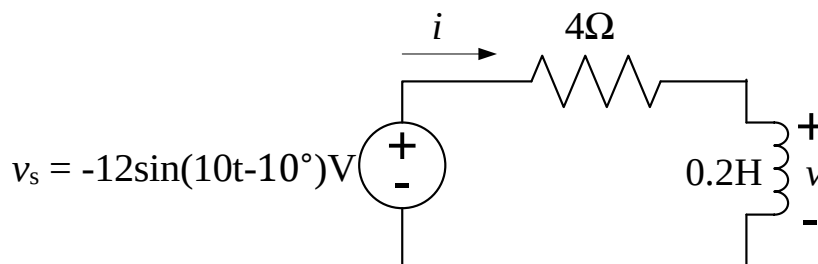
EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Final Examination, Fall 2021 Semester

Course: CSE 209 Electrical Circuits, Section-3
Instructor: Musharrat Khan, Senior Lecturer, CSE Department
Full Marks: 40 (20 will be counted for final grading)
Time: 1 Hour and 25 Minutes (Including submission)

Note: There are FIVE questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

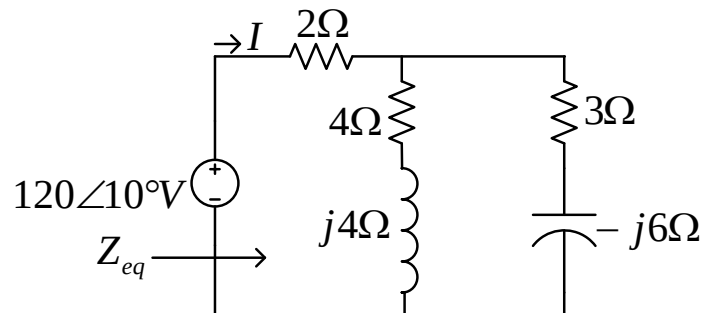
1. Determine $v(t)$ and $i(t)$ in the following circuit.

[CO1,C2,
Mark: 6]



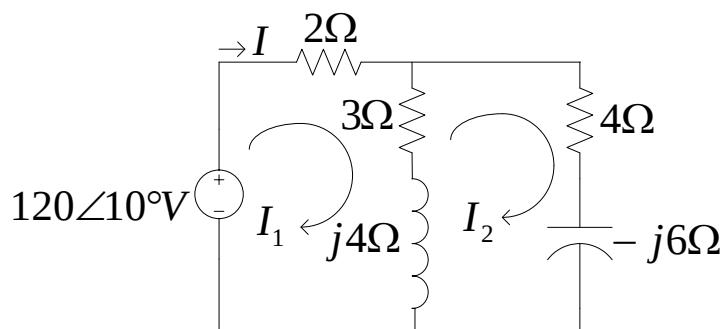
2. Determine Z_{eq} and I for the following circuit.

[CO1,C2,
Mark: 6]



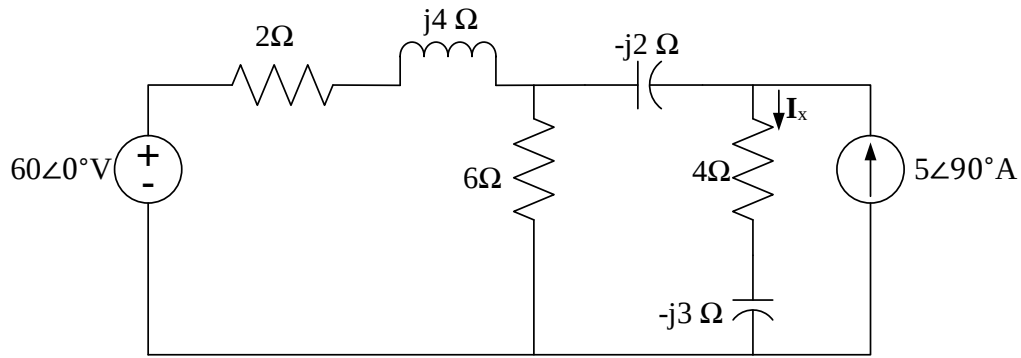
3. Using mesh currents indicated in the circuit, **compute** the appropriate mesh current for determining I in the following circuit.

[CO3,C4,
Mark: 8]



4. Using superposition theorem, **find** I_x in the following circuit.

[CO3,C4,
EP1,
Mark: 8]



5. **Find** the value of Z_L that will absorb the maximum power and the value of the maximum power in the following circuit.

[CO3,C4,
EP1,EP2,
Mark:12]

